MATH 281 Applied Boundary Value Problems

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1 Heat Equation

Definition 1.1. A **partial differential equation** (PDE) is an equation containing partial derivatives.

1.1 Conduction of Heat in a 1-D Rod

If we have 1 dimensional rod with cross sectional area A, L, and thermal energy density $e(x,t)$, the heat energy of a slice on the rod from x to $x+\Delta x$ is:

$$\text{heat energy} = e(x,t)A\Delta x$$

If there is no lateral heat flow on the rod, then the rate of change of the heat energy of this slice must be the sum of the net heat flux into the slice and the heat generated in the slice:

$$\frac{\partial}{\partial t}[e(x,t)A\Delta x] \approx \phi(x,t)A - \phi(x+\Delta x,t)A + Q(x,t)A\Delta X$$

Where $Q(x,t)$ is the heat energy generated per unit volume.

We can easily simplify this by diving by $A\Delta x$ and taking the $\lim_{\Delta x\to 0}$ to get the following PDE:

$$\frac{\partial e}{\partial t} = -\frac{\partial \phi}{\partial x} + Q$$

We can also derive this equation by considering a finite slice from $x=a$ to $x=b$. The rate of change of heat energy of this slice is the **integral conservation law**:

$$\frac{d}{dt}\int_a^b e dx = \phi(a,t) - \phi(b,t) + \int_a^b Q dx$$

We can convert this to a single integral by noting that $\frac{d}{dt}\int_a^b e(x,t) = \int_a^b \frac{\partial e(x,t)}{t}$ and that $\phi(a,t) - \phi(b,t) = -\int_a^b \frac{\partial \phi}{\partial x} dx$, resulting in:

$$\int_a^b (\frac{\partial e}{\partial t} + \frac{\partial \phi}{\partial x} - Q)dx = 0$$

Since this must be true for any arbitrary a and b, the integrand must be zero:

$$\frac{\partial e}{\partial t} = -\frac{\partial \phi}{x} + Q$$

Because we also describe the temperatures of materials instead of heat energy, it is useful to convert our equations. We let:

$$\begin{aligned} u(x,t) &\equiv \text{temperature} \\ c &\equiv \text{specific heat capacity} \\ \rho &\equiv \text{density} \end{aligned}$$

Then, the thermal energy density can be expressed as:

$$e(x,t) = c(x)\rho(x)u(x,t)$$

Therefore, the conservation of thermal energy can be expressed as:

$$c(x)\rho(x)\frac{\partial u}{\partial t} = -\frac{\partial \phi}{\partial x} + Q$$

Law 1.1. Fourier's law of heat conduction

$$\phi = -K_0 \frac{\partial u}{\partial x}$$

where K_0 is the thermal conductivity of the material.

Substituting this in, we get:

$$c\rho\frac{\partial u}{\partial t} = \frac{\partial}{\partial x}(K_0\frac{\partial u}{\partial x}) + Q$$

If c, ρ, and K_0 are all constant, then this simplifies to the **heat equation**:

$$\frac{\partial u}{\partial t} = k\frac{\partial^2}{\partial x^2}$$

where

$$k = \frac{K_0}{c\rho}$$

is the **thermal diffusivity**.

We can similarly derive integral and regular diffusion laws for diffusion of a chemical:

$$\frac{d}{dt}\int_a^b u(x,t)dx = \phi(a,t) - \phi(b,t)$$

$$\frac{\partial u}{\partial t} + \frac{\partial \phi}{\partial x} = 0$$

where $u(x,t)$ is the density or concentration of the chemical per unit volume.

Law 1.2. Fick's Law of Diffusion

$$\phi = -k\frac{\partial u}{\partial x}$$

where k is chemical diffusivity.

Using this, we get the **diffusion equation**:

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

1.2 Initial and Boundary Conditions

Since the heat equation has one time derivative, we need one **initial condition** (IC). Additionally, we need one **boundary condition** (BC) at each end of the rod.

One potential boundary condition is having a set temperature, called the **prescribed temperature**, at a boundary:

$$u(0,t) = u_B(t)$$

Definition 1.2. A **Dirichlet condition** is one where the values of a function at a boundary are specified.

We can also specify heat flow at a boundary:

$$-K_0(0)\frac{\partial u}{\partial x}(0,t) = \phi(t)$$

The simplest case of this is where the boundary is insulated:

$$\frac{\partial u}{\partial x} = 0$$

Definition 1.3. A **Neumann condition** is one where the derivative $\frac{\partial u}{\partial x}$ at a boundary is specified.

Finally, we can allow the boundary to exchange heat with the surrounding.

Theorem 1.1. Newton's Law of Cooling

The rate of heat transfer between a body and its surroundings can be approximated by the following:

$$-K_0(0)\frac{\partial u}{\partial x}(0,t) = -H[u(0,t) - u_B(t)]$$

where H is called the **heat transfer coefficient**.

1.3 Equilibrium Temperature Distribution

1.3.1 Prescibed Temperatures

Definition 1.4. An **equilibrium** or **steady-state** solution to a partial differential equation is one that does not depend on time.

Suppose we have a simplified scenario where the thermal conditions are constant, no heat is generated, and heat is prescribed on the boundaries:

$$u(0) = T_1$$
$$u(L) = T_2$$

We can easily found the equilibrium solution by setting the rate of heat flow to zero, double integrating, and solving:

$$\frac{d^2u}{dx^2} = 0$$
$$\to u(x) = C_1x + C_2$$

We then use the boundary conditions to solve for C_1 and C_2:

$$u(0) = C_2 = T_1$$
$$u(L) = C_1L + T_2 \to C_1 = \frac{T_2 - T_1}{L}$$

Thus, we have the following solution:

$$u(x) = T_1 + \frac{T_2 - T_1}{L}x$$

1.3.2 Insulated Boundaries

If instead the boundaries are insulated, we have the following PDE:

$$\begin{aligned} \text{PDE:} \quad & \frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} \\ \text{IC:} \quad & u(x,0) = f(x) \\ \text{BC1:} \quad & \frac{\partial u}{\partial x}(0,t) = 0 \\ \text{BC2:} \quad & \frac{\partial u}{\partial t}(L,t) = 0 \end{aligned}$$

We similarly solve for the equilibrium solution:

$$\begin{aligned} \text{ODE:} \quad & \frac{d^2u}{dx^2} = 0 \\ \text{BC1:} \quad & \frac{du}{dx}(0) = 0 \\ \text{BC2:} \quad & \frac{du}{dx}(L) = 0 \\ & \to u = C_1x + C_2 \to C_1 = 0 \to u = C_2 \end{aligned}$$

Since the boundaries are insulated, the total energy in the rod must be constant, so:

$$\int_0^L c\rho u dx = \int_0^L c\rho f(x)$$
$$\to u(x) = C_2 = \frac{1}{L}\int_0^L f(x)dx$$

2 Method of Separation of Variables

2.1 Linearity

Definition 2.1. An operator L is **linear** (and thus called a **linear operator**) if:

$$L(c_1u_1 + c_2u_2) = c_1L(u_1) + c_2L(u_2)$$

for any functions u_1 and u_2 and constants c_1 and c_2.

Definition 2.2. A **linear equation** for the unknown u is of the form:

$$L(u) = f$$

where L is a linear operator and f is known.

Definition 2.3. A linear equation $L(u) = f$ is called **homogeneous** if $f = 0$. Otherwise, it is called **nonhomogeneous**.

Theorem 2.1. Principle of Superposition
If u_1 and u_2 satisfy an arbitrary linear homogenous equation, then any arbitrary combination $c_1u_1 + c_2u_2$, where c_1 and c_2 are constants, also satisfies the same equation.

2.2 Heat Equation with Zero Temperature at Finite Ends

Suppose we have a rod with fixed temperatures of 0 at the boundaries, represented by the following:

$$\begin{aligned} \text{PDE:} \quad & \frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} \quad 0 < x < L, \quad t > 0 \\ \text{BC:} \quad & u(0,t) = 0 \\ & u(L,t) = 0 \\ \text{IC:} \quad & u(x,0) = f(x) \end{aligned}$$

We can solve this using the **method of separation of variables**.

2.2.1 Product Solutions

The first step is to determine a **product solution** of the form:

$$u(x,t) = H(x)G(t)$$

Substituting this into the PDE and dividing by $kH(x)G(t)$ to separate variables gives us:

$$\frac{1}{kG}\frac{dG}{dt} = \frac{1}{H}\frac{d^2H}{dH^2}$$

For this to be true for any x and t, both sides must equal a constant, which we set as $-\lambda$, where λ is the **separation constant**. This gives us two ordinary differential equations, one for $H(x)$ and one for $G(t)$.

The product solutions must satisfy the boundary conditions. $u(0,t) = 0$ implies that either $H(0) = 0$ or $G(t) \equiv 0$. The latter is uninteresting and is called the **trivial solution**, and thus:

$$\begin{aligned} H(0) &= 0 \\ H(L) &= L \end{aligned}$$

2.2.2 Time-Dependent Ordinary Differential Equation

We first solve the ODE for $G(t)$:

$$\frac{dG}{dt} = -\lambda k G$$

Since this is a first-order linear homogeneous ODE with constant coefficients, the solution is an exponential of the form:

$$G(t) = ce^{-\lambda k t}$$

2.2.3 Boundary Value (Eigenvalue) Problem

Next, we solve the second-order ODE for $H(x)$:

$$\begin{aligned} \frac{d^2H}{dx^2} &= -\lambda H \\ H(0) &= 0 \\ H(L) &= 0 \end{aligned}$$

Since there are boundary conditions, this is called a **boundary value problem**.
We try three different scenarios for the **eigenvalue** λ:

- $\lambda < 0$
  If we let $\lambda = -\beta^2$, our ODE has solutions of the form:

  $$H(x) = c_1 \cosh(\beta x) + c_2 \sinh(\beta x)$$

  We then apply the boundary conditions:

  $$\begin{aligned} H(0) &= c_1 = 0 \\ H(L) &= c_2 \sinh \beta L = 0 \to c_2 = 0 \quad \text{(if } \beta \text{ is positive)} \end{aligned}$$

  Thus, this is trivial and there are no negative eigenvalues.

- $\lambda = 0$
  If $\lambda = 0$, we can simply integrate twice and find that solutions are of the form:

  $$H(x) = c_1 x + c_2$$

  We then apply the boundary conditions:

  $$\begin{aligned} H(0) &= c_2 = 0 \\ H(L) &= c_1 L = 0 \to c_1 = 0 \end{aligned}$$

  Thus, this is also trivial and there is no eigenvalue of zero.

- $\lambda > 0$
  If we let $\lambda = \alpha^2$, then the ODE has solutions of the form:

  $$H(x) = c_1 \cos(\alpha x) + c_2 \sin(\alpha x)$$

  We then apply the boundary conditions:

  $$\begin{aligned} H(0) &= c_1 = 0 \\ H(L) &= c_2 \sin \alpha L = 0 \to \alpha L = n\pi \quad \text{for the non-trivial case} \end{aligned}$$

  Thus, the eigenvalues are:

  $$\lambda = (\frac{n\pi}{L})^2, \quad n = 1, 2, 3, \ldots$$

  And the corresponding **eigenfunctions** are:

  $$H(x) = \sin \frac{n\pi x}{L}$$

2.2.4 Fourier Series

The overall solution of our PDE is:

$$u(x,t) = \sum_{n=1}^{m} B_n \sin\frac{n\pi x}{L} e^{\frac{-ktn^2\pi^2}{L^2}}$$

Next, we need to find B_n using the initial condition $f(x)$. In some situations this is easy, but for many functions, it isn't possible if m is finite and requires a **Fourier series**:

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L}$$

We know that:

$$\int_0^{\frac{L}{2}} \sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}dx = \begin{cases} 0 & m \neq n \\ \frac{L}{2} & m = n \end{cases}$$

Thus, we can multiply both sides by $\sin\frac{m\pi x}{L}$, integrate, and rearrange to get:

$$\begin{aligned} f(x)\sin\frac{m\pi x}{L} &= \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L} \\ \rightarrow \int_0^L f(x)\sin\frac{m\pi x}{L}dx &= \int_0^L \sum_{n=0}^{\infty} B_n \sin\frac{n\pi x}{L}\frac{m\pi x}{L}dx \\ \rightarrow \int_0^L f(x)\sin\frac{m\pi x}{L}dx &= \frac{L}{2}B_m \\ \rightarrow B_m &= \frac{2}{L}\int_0^L f(x)\sin\frac{m\pi x}{L}dx \end{aligned}$$

Since functions are vectors, this is essentially taking the inner product of $f(x)$ and $\sin\frac{m\pi x}{L}$:

$$\langle f(x), \sin\frac{m\pi x}{L}\rangle = \int_0^L f(x)\sin\frac{m\pi x}{L}dx$$

Definition 2.4. Two functions $f_1(x)$ and $f_2(x)$ are **orthogonal** over the interval $[a,b]$ if:

$$\langle f_1, f_2\rangle = \int_a^b f_1 f_2 dx = 0$$

Since functions of the form $\sin(nx)$ are orthogonal if n is different, the eigenfunctions of our solution are an **orthogonal set of vectors**.

3 Fourier Series

Definition 3.1. A **Fourier series** is an infinite series of the form:

$$f(x)\ a_0 + \sum_{n=1}^{\infty} a_N \cos\frac{n\pi}{L}x + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi}{L}x$$

Definition 3.2. A function $f(x)$ is defined to be **piecewise smooth** on some interval if the interval can be divided into a finite number of subintervals such that

1. Both $f(x)$ and $f'(x)$ are continuous at all points on the interior of the subintervals
2. The one-sided limits of $f(x)$ and $f'(x)$ both exist (as finite values) at the boundaries of the subintervals

3.0.1 Statement of Convergence Theorem

Fourier coefficients can be found using the following equations:

$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)dx$$
$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}dx$$
$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}dx$$

Theorem 3.1. Convergence Theorem (Fourier's Theorem)
If $f(x)$ is piecewise smooth on the interval $-L \le x \le L$, then the Fourier series of $f(x)$ converges for all $-\infty < x < \infty$.
Let $f_E(x)$ denote the **periodic extension** of $f(x)$.

1. At any point x where $f_E(x)$ is continuous, the Fourier series of $f(x)$ converges to $f_E(x)$.
2. At any point x where $f_E(x)$ has a jump discontinuity, the Fourier series of $f(x)$ converges to the average of the left
3. and right hand limits:
$$\frac{1}{2}(f_E(x+) + f_E(x-))$$

3.1 Fourier Cosine and Sine Series

3.1.1 Fourier Cosine Series

For even functions, the sine coefficients of the Fourier series are zero. If we have a function $f(x)$ defined on $0 < x < L$, we can create an *evenextension* of $f(x)$ that, as one might guess, is an even function. Since the Fourier series of an even function only has cosine terms, it is called a **Fourier cosine series**:

$$f(x)\ \sum_{n=0}^{\infty} A_n \cos\frac{n\pi x}{L}, \quad 0 \le x \le L$$
$$A_0 = \frac{1}{L}\int_{0}^{L} f(x)dx$$
$$A_n = \frac{2}{L}\int_{0}^{L} f(x)\cos\frac{n\pi x}{L}dx$$

3.1.2 Fourier Sine Series

For odd functions, the cosine coefficients of the Fourier series are zero. If we have a function $f(x)$ defined on $0 \le x \le L$, we can create an **odd extension** of $f(x)$ that, as the name implies, is an odd function. Since the Fourier series of an odd function only has sine terms, it is called a **Fourier sine series**:

$$f(x)\ \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L}, \quad 0 \le x \le L$$
$$B_n = \frac{2}{L}\int_{0}^{L} f(x)\sin\frac{n\pi x}{L}dx$$

3.1.3 Even and Odd Parts

Any function $f(x)$ can be decomposed into a combination of an even function, called the **even part** of $f(x)$, and an odd function, called the **odd part** of $f(x)$:

$$f(x) = f_e(x) + f_o(x)$$
$$f_e(x) = \frac{1}{2}[f(x) + f(-x)]$$
$$f_o(x) = \frac{1}{2}[f(x) - f(-x)]$$

Thus, the Fourier series of some function $f(x)$ is equal to the Fourier sine series of $f_o(x)$ plus the Fourier cosine series of $f_e(x)$.

3.1.4 Continuous Fourier Series

For a piecewise smooth function $f(x)$, the Fourier series of $f(x)$ is continuous and converges to $f(x)$ for $-L \leq x \leq L$ if and only if $f(x)$ is continuous and $f(-L) = f(L)$.

For a piecewise smooth function $f(x)$, the Fourier cosine series of $f(x)$ is continuous and converges to $f(x)$ for $0 \leq x \leq L$ if and only if $f(x)$ is continuous.

For a piecewise smooth function $f(x)$, the Fourier sine series is continuous and converges to $f(x)$ for $0 \leq x \leq L$ if and only if $f(x)$ is continuous and both $f(0) = 0$ and $f(L = 0)$.

The **Gibbs phenomenon** is an example of overshoot that occurs at jump discontinuities, with an overshoot of around 9%. It occurs when a finite series of eigenfunctions approximates a discontinuous function.

4 Term-By-Term Differentiation of Fourier Series

If $f'(x)$ is piecewise smooth, then the Fourier series of a continuous function $f(x)$ can be differentiated term by term if $f(-L) = f(L)$.

If $f'(x)$ is piecewise smooth, then the Fourier cosine series of a continuous function $f(x)$ can be differentiated term by term.

If $f'(x)$ is piecewise smooth, then the Fourier sine series of a continuous function $f(x)$ can be differentiated term by term only if $f(0) = 0$ and $f(L) = 0$. ‘’

If $f'(x)$ is piecewise smooth, then the Fourier sine series of a continuous function $f(x)$,

$$f(x) \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L}$$

cannot, in general, be differentiated term by term. However:

$$f'(x) \frac{1}{L}[f(L) - f(0)] + \sum_{n=1}^{\infty}[\frac{n\pi}{L}B_n + \frac{2}{L}((-1)^n f(L) - f(0))] \cos \frac{n\pi x}{L}$$

5 Method of Eigenfunction Expansion

Suppose we have the heat equation with boundaries of $u = 0$ at $x = 0$ and $x = L$.

We start with the assumption that $u(x, t)$ is continuous and that $\frac{\partial u}{\partial t}$, $\frac{\partial u}{\partial x}$, and $\frac{\partial^2 u}{\partial x^2}$ are also continuous.

6 Numerical Solutions of PDEs

6.1 Linear Advection Equation

Consider the following linear PDE, called the **advection equation**:

$$\frac{\partial}{\partial t}u(x, t) + U_0 \frac{\partial}{\partial x}u(x, t) = 0$$

We note that the total time derivative is:

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \frac{dx}{dt}\frac{\partial}{\partial x}$$

Therefore, if $\frac{dx}{dt} = U_0$, our PDE is just :

$$0 = \frac{\partial}{\partial t}u(x,t) + \frac{dx}{dt}\frac{\partial}{\partial x}u(x,t) = \frac{d}{dt}u$$

This is effectively saying that u is moving at speed U_0, and within its frame of reference (along paths given by $\frac{d}{dt}x$), it is constant.

We easily determine that $x(t) = x_0 + U_0 t$ and thus $x_0 = x(t) - U_0 t$, meaning we can convert from a point (x,t) to a point $(x_0, 0)$ with the same value of u and thus that we've reduced our solution to:

$$u(x(t),t) = u(x - U_0 t, 0) = u_0(x - U_0 t)$$

6.1.1 Burger's Equation

The following non-linear PDE is known as **Burger's equation**:

$$\frac{\partial}{\partial t}u + u\frac{\partial}{\partial x}u = 0$$

6.1.2 Introduction to Linearization

Since Burger's Equation is non-linear, it's difficult to solve analytically, but we can approximate it using a technique called **linearization**.

We approximate u with a **reference state** and a **deviation**:

$$\underbrace{u(x,t)}_{\text{Full solution}} = \underbrace{U_0}_{\text{Reference state (constant)}} + \underbrace{\epsilon}_{\text{"Small constant"}} * \underbrace{\tilde{u}(x,t)}_{\text{Perturbation}}$$

Substituting this into the PDE and noting that U_0 is constant and a solution to the PDE gives us:

$$\frac{\partial}{\partial t}(U_0 + \epsilon\tilde{u}) + (U_0 + \epsilon\tilde{u})\frac{\partial}{\partial x}(U_0 + \epsilon\tilde{u}) = 0$$

$$\rightarrow \underbrace{\frac{\partial}{\partial t}U_0 + U_0\frac{\partial}{\partial x}U_0}_{U_0 \text{ solution to PDE}}{}^{\nearrow 0} + \epsilon\frac{\partial}{\partial t}\tilde{u} + \underbrace{\epsilon\tilde{u}\frac{\partial u}{\partial x}U_0}_{U_0 \text{ constant}}{}^{\nearrow 0} + U_0\epsilon\frac{\partial}{\partial x}\tilde{u} + \epsilon^2\tilde{u}\frac{\partial}{\partial x}\tilde{u} = 0$$

$$\rightarrow \epsilon[\frac{\partial}{\partial t}\tilde{u} + U_0\frac{\partial}{\partial x}\tilde{u} + \epsilon\tilde{u}\frac{\partial}{\partial x}\tilde{u}] = 0$$

$$\rightarrow \frac{\partial}{\partial t}\tilde{u} + U_0\frac{\partial}{\partial x}\tilde{u} + \underbrace{\epsilon\tilde{u}\frac{\partial}{\partial x}\tilde{u}}_{\text{non-linear}} = 0$$

If $|\epsilon| \ll 1$, then we have the advection equation:

$$\frac{\partial}{\partial t}\tilde{u} + U_0\frac{\partial}{\partial x}\tilde{u} = 0$$

6.2 Numerical Methods

It is often difficult or impossible to analytically solve non-linear PDEs, and thus we use numerical methods to approximate their solutions. However, numerical methods have the following drawbacks:

- A numerical solution only applies to one set of initial conditions, unlike analytical solutions that apply everywhere
- Numerical solutions (typically) have inherent error, so we need to understand it and keep it within a reasonable range
- Sometimes the error isn't large enough to be glaringly obvious, but still breaks the laws of physics, so numerical methods should be paired with analytical theory like checking conservation of energy.

6.2.1 Notation

Suppose $x \in [a, b]$ and $t \geq t_0$. We discretize x and t and let x_j be the jth point in space and t_n be the nth point in time. Then:

$$u_j^n = u(x_j, t_n)$$

Our spatial grid is $\{x_0, x_1, \ldots, x_{N-1}\}$ where N is the total number of points in our grid. Since $x \in [a, b]$, we have $a \leq x_0$ and $x_{N-1} \leq b$ (need not be strict equalities), and our points are ordered such that $i < j \to x_i < x_j$. The spacing does not have to be uniform but we assume it is to make our math easier.

6.2.2 General Procedure

Suppose we have the following PDE:

$$\partial_t u + u\partial_x u = 0$$

with initial conditions:

$$u(x_i, 0) = u_0(x_i)$$

We go through the following steps:

1. Use a **spatial scheme** to approximate $\partial_x u$ at each x_i. Then, we let:

$$F_i = u_i \frac{\partial}{\partial x} u|_{x=x_i}$$

2. Use a **temporal scheme** on the resulting ODEs:

$$\frac{d}{dt} u_i = -F_i$$

   to advance the solution forward in time from t_0 to some time $t_0 + \Delta t$

3. Repeat until desired end time is reached.

6.2.3 Spatial Scheme

Finite Difference (FD) methods use Taylor expansions to find derivatives. If we have u_i, we can find u at nearby points:

$$\begin{aligned}
u_{i-2} &\approx u_i - 2\Delta x \frac{\partial u}{\partial x}|_{x=x_i} + \frac{1}{2} 4\Delta x^2 \frac{\partial^2 u}{\partial x^2}|_{x=x_i} + \mathcal{O}(\Delta x^3) \\
u_{i-1} &\approx u_i - \Delta x \frac{\partial u}{\partial x}|_{x=x_i} + \frac{1}{2} \Delta x^2 \frac{\partial^2 u}{\partial x^2}|_{x=x_i} + \mathcal{O}(\Delta x^3) \\
u_i &= u_i \\
u_{i+1} &\approx u_i + \Delta x \frac{\partial u}{\partial x}|_{x=x_i} + \frac{1}{2} \Delta x^2 \frac{\partial^2 u}{\partial x^2}|_{x=x_i} + \mathcal{O}(\Delta x^3) \\
u_{i+2} &\approx u_i + 2\Delta x \frac{\partial u}{\partial x}|_{x=x_i} + \frac{1}{2} 4\Delta x^2 \frac{\partial^2 u}{\partial x^2}|_{x=x_i} + \mathcal{O}(\Delta x^3)
\end{aligned}$$